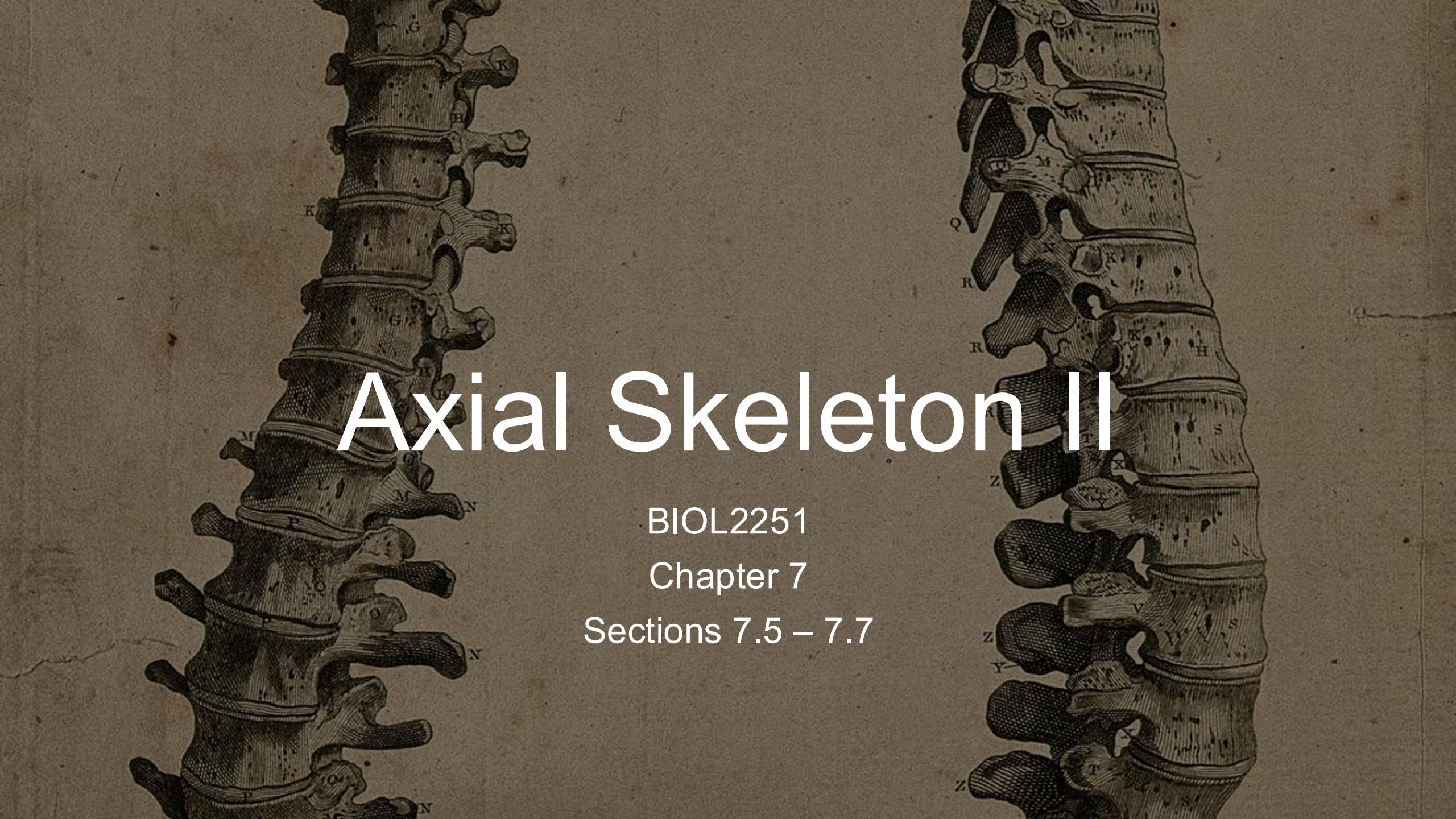




Axial Skeleton II

BIOL2251

Chapter 7

Sections 7.5 – 7.7

Vertebral Column Brainstorm



Vertebral Column Brainstorm

- **Instructions**

- Think on your own for 90 seconds
 - **List as many functions of the vertebral column as possible**
- Compare your work with a partner
- Share your answers I compile a list on the whiteboard
- **Bloom's taxonomy = Remember**

Vertebral Column Brainstorm

- Supports the weight of the head, neck, and trunk and transfers that load to the lower limbs
- Protects the spinal cord within the vertebral canal
- Provides attachment points for muscles of the back, neck, and thorax
- Allows movement: flexion, extension, lateral flexion, and rotation
- Serves as an attachment site for the ribs (via thoracic vertebrae)
- Maintains upright posture and the characteristic spinal curves that absorb shock



Spinal Curve Sketching

Spinal Curve Sketching

- **Instructions**

- Draw a basic lateral outline of the vertebral column
 - Label the four regions and indicate if they are primary or secondary
 - Note if the curve opens anteriorly or posteriorly
 - Volunteers will draw and label on the large whiteboard
- **Bloom's taxonomy = Remember, Understand**

Spinal Curve Sketching

Spinal Curves

Primary curves develop before birth, and secondary curves after birth.

The **cervical curve**, a secondary curve, develops as the infant learns to balance the weight of its head on the vertebrae of the neck.

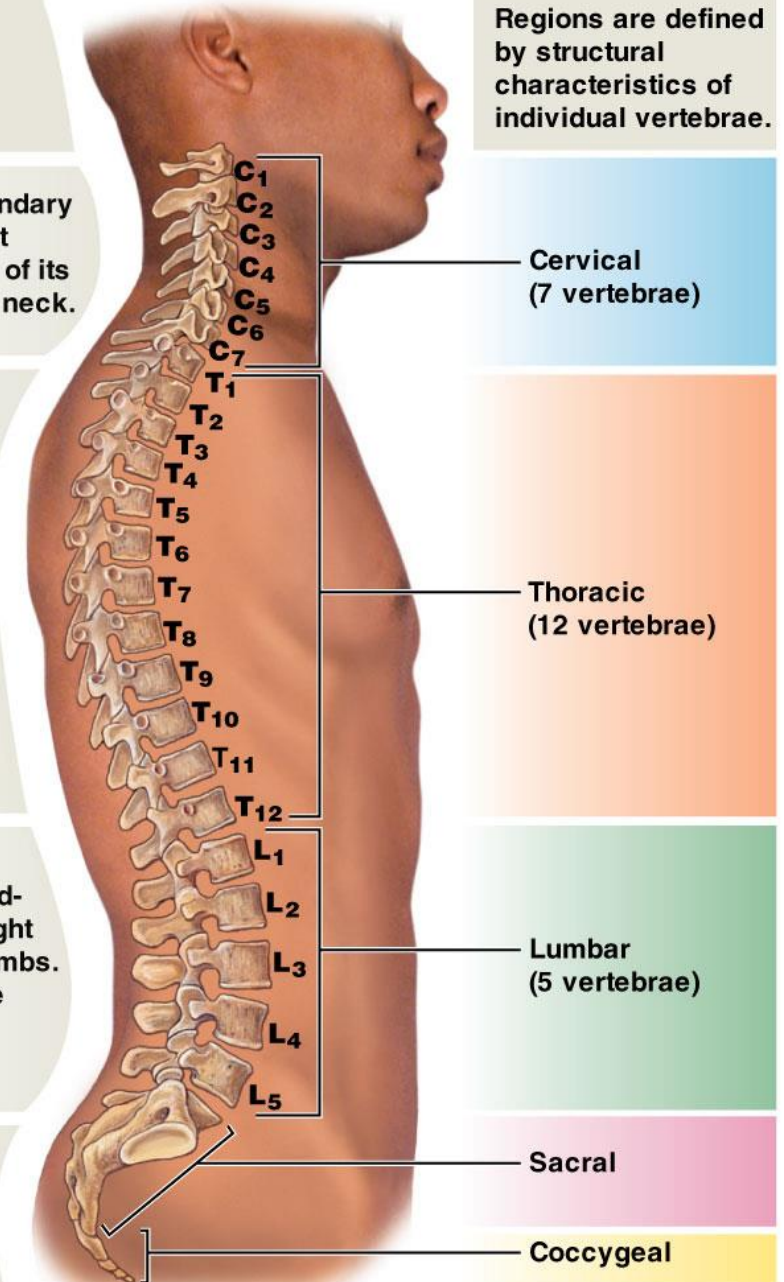
The **thoracic curve**, a primary curve, provides room for the thoracic organs.

The **lumbar curve**, a secondary curve, balances the weight of the trunk over the lower limbs. This curve develops with the ability to stand.

The **sacral curve**, a primary curve, provides room for various abdominopelvic organs.

Vertebral Regions

Regions are defined by structural characteristics of individual vertebrae.



Spinal Curve Sketching

Region	Curve Direction	Primary or Secondary	Notes
Cervical	Anteriorly convex	Secondary	Develops when infant begins lifting head
Thoracic	Posteriorly convex	Primary	Present at birth; retained fetal curve
Lumbar	Anteriorly convex	Secondary	Develops when child begins standing/walking
Sacral	Posteriorly convex	Primary	Present at birth; retained fetal curve



Curve Case Identification

Curve Case Identification

- **Instructions**

- Work in small groups (4-6 people)
- I will project 3 brief cases
 - **Identify the spinal curve abnormality in each case and describe what is abnormal about the curve**
- Groups share answers before whole-class review

- **Bloom's taxonomy = Understand, Apply**

Curve Case Identification

- **Case A:** A 70-year-old woman presents with a pronounced forward rounding of the upper back. She appears to be hunching forward and has lost 2 inches of height over the past decade. Lateral imaging shows an exaggerated posterior convexity in the thoracic region.
- **Case B:** A 45-year-old man with obesity presents with chronic lower back pain. On lateral examination, his abdomen protrudes significantly and his lower back has an exaggerated inward curve.
- **Case C:** A 14-year-old girl is screened during a school physical. When she bends forward, her right shoulder blade appears higher than the left. Posterior imaging shows a lateral S-shaped deviation of the spine.

Curve Case Identification

- Case A: **Kyphosis** — an exaggerated posterior convexity of the thoracic curve, commonly associated with osteoporosis-related vertebral compression fractures in older adults.
- Case B: **Lordosis** — an exaggerated anterior convexity of the lumbar curve. Excess abdominal weight shifts the center of gravity forward, causing compensatory lumbar hyperextension.
- Case C: **Scoliosis** — an abnormal lateral curvature of the spine, often with a rotational component. Adolescent idiopathic scoliosis is the most common form and is frequently caught during school screenings.



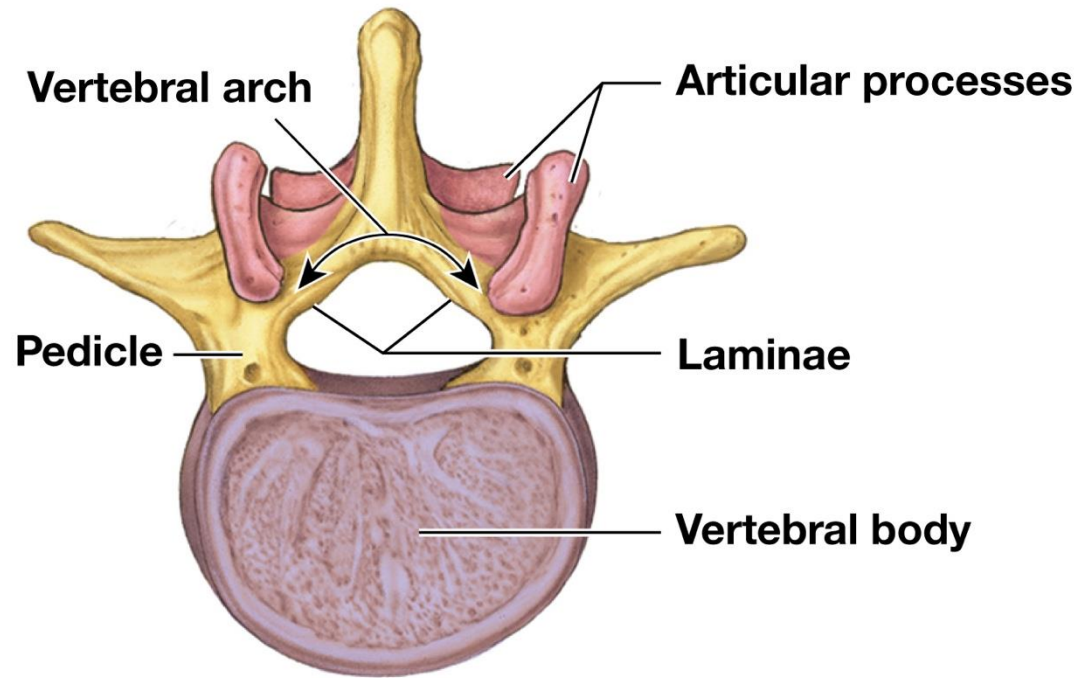
Build-a-Vertebra

Build-a-Vertebra

- **Instructions**

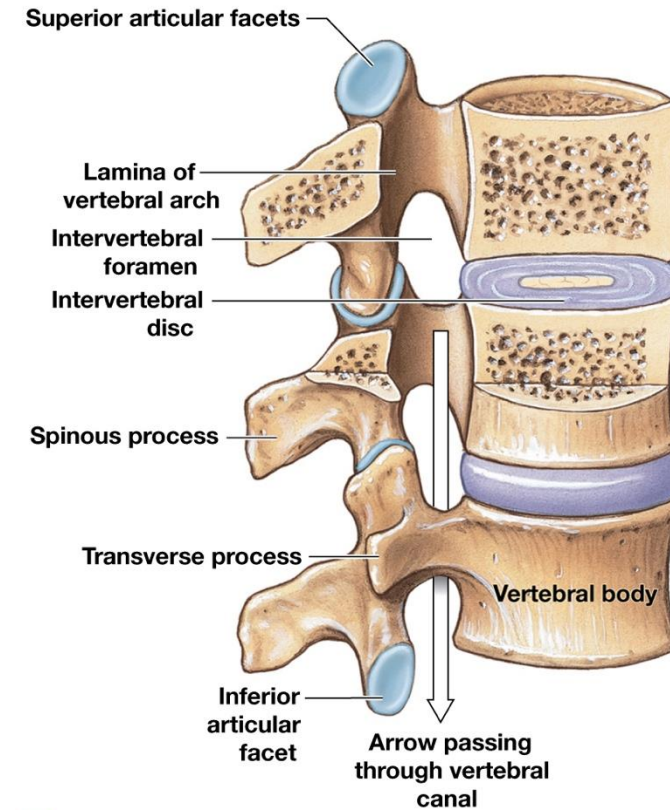
- Work in small groups (12 total)
- Draw a **superior view of a generic vertebrae** and label the following structures:
 - **Vertebral body, vertebral foramen, spinous process, transverse process, superior articular processes, and articular facets**
- Draw a **lateral view** and indicate the location of the following structures:
 - **Intervertebral disc, intervertebral foramen**
- Review the structure as a class
- **Bloom's taxonomy = Remember, Understand**

Build-a-Vertebra



a The major parts of a typical vertebra

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e A lateral and sectional view of three articulated vertebrae

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Build-a-Vertebra

Superior View Labels

- Vertebral body — large, anterior, cylindrical or kidney-shaped; weight-bearing portion
- Vertebral foramen — opening posterior to the body through which the spinal cord passes; together all foramina form the vertebral canal
- Spinous process — single posterior projection; palpable along the midline of the back
- Transverse processes — paired lateral projections; muscle and ligament attachment
- Superior articular processes — paired projections facing upward; bear articular facets that articulate with the inferior facets of the vertebra above
- Articular facets — smooth cartilage-covered surfaces on the articular processes where adjacent vertebrae contact each other

Lateral View Labels

- Intervertebral disc — fibrocartilage pad sitting between adjacent vertebral bodies; shock absorption and spacing
- Intervertebral foramen — opening formed between the pedicles of adjacent vertebrae (not within a single vertebra); where spinal nerves exit the vertebral canal

Process Matching



Process Matching

- **Instructions**

- Work on your own
 - Feel free to discuss with a partner
- I will project a table with structures and descriptions
 - **Match the name of the structure to the correct description**
- We will debrief as a class

- **Bloom's taxonomy = Understand, Analyze**

Process Matching

Description

- The lateral opening formed between the pedicles of two adjacent vertebrae; the exit point for spinal nerves
- Paired lateral projections extending to each side of the vertebral arch; serve as muscle and ligament attachment sites
- The smooth, flat cartilage-covered surface on an articular process; its orientation determines what movements are possible in that spinal region
- A fibrocartilage pad sitting between adjacent vertebral bodies; composed of a gel-like center surrounded by a tough fibrous ring

Structure

- A. Articular facet
- B. Articular process
- C. Intervertebral disc
- D. Intervertebral foramen
- E. Spinous process
- F. Transverse process
- G. Vertebral body
- H. Vertebral foramen

Process Matching

Description

- The large, anterior, cylindrical portion of the vertebra; increases in size from cervical to lumbar reflecting increasing weight load
- The single unpaired projection extending posteriorly from the vertebral arch; palpable along the midline of the back
- The enclosed space within a single vertebra formed between the vertebral body and vertebral arch; collectively all of these form the vertebral canal
- Paired superior and inferior projections arising from the vertebral arch that bear smooth surfaces; link adjacent vertebrae into joints

Structure

- A. Articular facet
- B. Articular process
- C. Intervertebral disc
- D. Intervertebral foramen
- E. Spinous process
- F. Transverse process
- G. Vertebral body
- H. Vertebral foramen

Process Matching

Description	Structure
The lateral opening formed between the pedicles of two adjacent vertebrae; the exit point for spinal nerves	Articular facet
Paired lateral projections extending to each side of the vertebral arch; serve as muscle and ligament attachment sites	Articular process
The smooth, flat cartilage-covered surface on an articular process; its orientation determines what movements are possible in that spinal region	Intervertebral disc
A fibrocartilage pad sitting between adjacent vertebral bodies; composed of a gel-like center surrounded by a tough fibrous ring	Intervertebral foramen
The large, anterior, cylindrical portion of the vertebra; increases in size from cervical to lumbar reflecting increasing weight load	Spinous process
The single unpaired projection extending posteriorly from the vertebral arch; palpable along the midline of the back	Transverse process
The enclosed space within a single vertebra formed between the vertebral body and vertebral arch; collectively all of these form the vertebral canal	Vertebral body
Paired superior and inferior projections arising from the vertebral arch that bear smooth surfaces; link adjacent vertebrae into joints	Vertebral foramen

Regional Vertebra Sorting



Regional Vertebra Sorting

- **Instructions**

- Work on your own (to start)
- I will read out a series of descriptions
- For each description, indicate if this is a cervical, thoracic, or lumbar vertebra
- Check with your neighbor, then we will review as a class

- **Bloom's taxonomy = Analyze**

Regional Vertebra Sorting

Description	Answer
Possess a bifid spinous process	Cervical
Contain a foramen within each transverse process for passage of the vertebral artery	Cervical
Have transverse foramina	Cervical
Spinous process projects posteriorly and somewhat inferiorly	Cervical
Have both costal facets on the body and transverse costal facets on the transverse process	Thoracic
Long, thin spinous process that projects posteriorly and inferiorly	Thoracic
Have facets for rib articulation	Thoracic
Spinous processes overlap the vertebra below like shingles on a roof	Thoracic
Circular, relatively small vertebral foramen	Thoracic
Largest vertebral bodies of the vertebral column to help with weight-bearing demand	Lumbar
Short, thick, blunt spinous processes that project posteriorly	Lumbar
Transverse processes are broad and serve as major muscle attachment sites; no foramina or rib facets present	Lumbar



Atlas and Axis —
Who Am I?

Atlas and Axis — Who Am I?

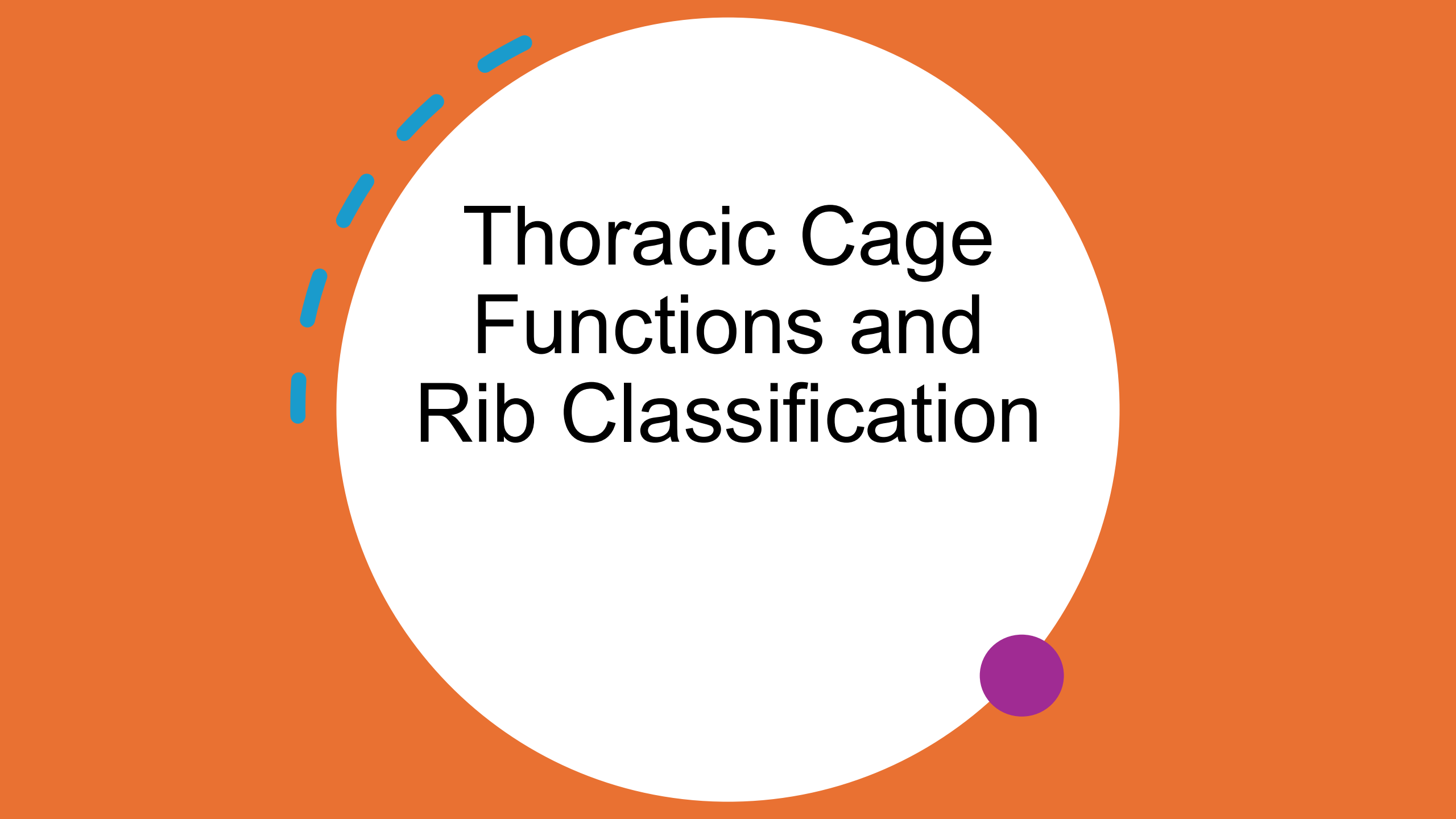
- **Instructions**

- Work in pairs or small groups
- I will read a clue about either the atlas or axis
 - **Based on the clue, identify if this describes the atlas or axis**
- Keep track of your work
- We will debrief as a class

- **Bloom's taxonomy = Remember, Understand**

Atlas and Axis — Who Am I?

Clue	Answer
I am named for the Titan who held the world on his shoulders — I hold the skull on top of the spine.	Atlas (C1)
I have no vertebral body and no spinous process — I am essentially a bony ring.	Atlas (C1)
I have two arches — one anterior and one posterior — connected by lateral masses.	Atlas (C1)
I am named for a pivot — I provide the structure that allows you to shake your head no.	Axis (C2)
I have a cylindrical projection that extends superiorly into the ring of C1, acting as a pivot point.	Dens (on C2)
I am the feature of C2 that was originally the vertebral body of C1 during development.	Dens (on C2)
The rotation that lets you shake your head no occurs between me and C1 — the dens of C2 pivots within the anterior arch of C1.	Axis (C2) / atlanto-axial joint
The nodding motion that lets you say yes occurs at the joint between C1 and the skull.	Atlas (C1) / atlanto-occipital joint



Thoracic Cage Functions and Rib Classification

Thoracic Cage Functions and Rib Classification

- **Instructions**

- Work in small groups (12 total)
- You will complete a two-part activity
- **Part A: list all the functions of the thoracic cage from memory**
 - Rank in order of most critical for survival
 - Be prepared to defend your ranking
- **Part B: Classify ribs 1-12 into true, false, and floating categories**
 - Explain the anatomical basis for each classification
- We will review (and maybe debate) as a class
- **Bloom's taxonomy = Remember, Understand Apply**

Thoracic Cage Functions and Rib Classification

Functions of the Thoracic Cage

- Protects the heart, lungs, large arteries and veins, liver (partially), and spleen
- Supports the weight of the pectoral girdle and upper limbs
- Provides attachment sites for muscles of the thorax, neck, upper limbs, and abdomen
- Enables breathing — the ribs move during inhalation and exhalation to change thoracic volume

Thoracic Cage Functions and Rib Classification

Classification	Ribs	Anatomical Basis
True ribs	1-7	Attach directly to the sternum via their own costal cartilage
False ribs	8-10	Costal cartilage does not reach the sternum directly — instead connects to the cartilage of rib 7
Floating ribs	11-12	Have no anterior attachment at all — costal cartilage is vestigial and does not connect to anything

Thoracic Cage Functions and Rib Classification

Rib Structure Key

- Head — posterior end of the rib; articulates with the costal facet on the vertebral body
- Tubercle — small projection near the head; articulates with the transverse costal facet on the transverse process of the thoracic vertebra
- Body (shaft) — the long, curved portion forming most of the rib
- Costal cartilage — anterior hyaline cartilage connecting the rib to the sternum (in true ribs) or to the rib above (in false ribs)



Sternum Sketch

Sternum Sketch

- **Instructions**

- Work on your own
- **Sketch the sternum from memory**
 - Label the manubrium, body, and xiphoid process
- Then answer the following clinical question:
 - During CPR, chest compressions are delivered to the lower half of the body of the sternum. Why is it important not to compress too inferiorly?
- We will review as a class
- **Bloom's taxonomy = Remember, Understand, Apply**

Sternum Sketch

- **Manubrium** — superior, broadest portion; articulates with the clavicles and the first pair of ribs; the jugular notch is the palpable superior depression at the base of the neck
- **Body** — long middle portion; articulates with ribs 2–7 via their costal cartilages; the junction between the manubrium and body creates a slight ridge called the sternal angle
- **Xiphoid process** — inferior, smallest portion; purely cartilaginous in youth, ossifies with age; attachment point for abdominal muscles and the diaphragm; has no rib attachments

Synthesis Voting (with Defense)



Synthesis Voting (with Defense)

- **Instructions**

- I will read statements aloud
 - **Vote thumbs up (true), thumbs down (false), or thumbs to the side (not sure)**
- After each vote, **I will ask a correctly-voting student to explain why**
- Student who explains their thought process = small prize

- **Bloom's taxonomy = Evaluate**

Synthesis Voting (with Defense)

- *The primary curves of the vertebral column are the cervical and lumbar curves.*
 - **FALSE**
 - The primary curves are the thoracic and sacral curves — they are retained from the original fetal C-shaped curvature and are present at birth. The cervical and lumbar curves are secondary curves that develop postnatally in response to muscle use and gravity.
- *Lumbar vertebrae have the largest vertebral bodies because they bear the most weight.*
 - **TRUE**
 - Weight load increases progressively down the vertebral column as each vertebra must support everything above it. Lumbar vertebrae carry the weight of the head, neck, thorax, and upper limbs, so their bodies are large and kidney-shaped to distribute that load effectively. This is a clear example of structure reflecting function.

Synthesis Voting (with Defense)

- *Costal facets are found on both the vertebral body and the transverse process of (most) thoracic vertebrae.*
 - **TRUE**
 - Most thoracic vertebrae have costal facets on the body (for articulation with the head of a rib) and transverse costal facets on the transverse process (for articulation with the tubercle of the same rib). This means each typical rib articulates with its thoracic vertebra at two points, creating a more stable but less mobile joint. T11 and T12 are exceptions — their ribs articulate only at the body, with no transverse costal facets.



Who Am I? More Bone Riddles

Who Am I? More Bone Riddles

- **Instructions**

- Work on your own
- I have a list of riddles to read aloud
 - **Each riddle indicates a bone, structure, or curve for you to identify**
- First person with their hand up gets the opportunity to share
- Correct identification = small prize

- **Bloom's taxonomy = Analyze**

Who Am I? More Bone Riddles

- I am one of two curves present in the spine at birth. I curve posteriorly — my convexity faces backward — and I am retained directly from the fetal position. I span the longest region of the vertebral column. I am considered a primary curve. Who am I?

→ Thoracic Curve

- I am an exaggerated posterior convexity of a spinal curve, most commonly seen in the thoracic region. I may cause a visible forward rounding of the upper back. In older adults I am often associated with vertebral compression fractures from osteoporosis. Who am I?

→ Kyphosis

Who Am I? More Bone Riddles

- I am an opening found within every individual vertebra, formed between the vertebral body in front and the vertebral arch behind. The spinal cord passes through me, and when all of my counterparts in every vertebra line up, we collectively form the vertebral canal. Who am I?

→ **Vertebral Foramen**

- I am one of a pair of projections extending laterally from the vertebral arch, one on each side. I serve as an attachment point for muscles and ligaments of the back. Depending on which vertebra I belong to, I may also carry a foramen for a major artery or a facet for a rib. Who am I?

→ **Transverse Process**

Who Am I? More Bone Riddles

- I am a feature found only in the cervical vertebrae. I am a small opening within each transverse process, and my primary role is to protect the vertebral artery as it ascends toward the brain. Even C7 has me, though I typically carry only veins at that level. Who am I?

→ **Transverse Foramina**

- We are the first seven pairs of ribs. Each of us has our own dedicated piece of costal cartilage that connects us directly and independently to the sternum. We are called 'true' because of this direct, individual sternal connection. Who are we?

→ **True Ribs**